

# P03/3-RS485-GPS, P03/3-RS485-CET Weather Station

## Technical specifications and installation instructions

Item numbers  
30145 (P03/3-RS485-GPS)  
30151 (P03/3-RS485-CET)



## 1. Description

The **P03/3-RS485-GPS/CET Weather Station** measures temperature, wind speed and brightness (eastern, southern and western sun). It recognizes precipitation and receives the time signal with a GPS receiver. With the weather station P03/3-RS485-GPS the international time signal UTC and position coordinates are output. Direction (azimuth) and hight (elevation) of the sun are also calculated and output. For the P03/3-RS485-CET Weather Station, the Central European Time (CET) is output. The daylight savings time is adjusted automatically according to the specifications for Central Europe.

The weather station sends the currently recorded weather data, date and time once every second. This data flow can be received and analyzed by an end device such as SPS, PC or MC.

### Functions:

- **Brightness measurement** with three separate sensors for east, south and west. Recognition of twilight/dawn
- **Wind measurement:** The wind strength measurement takes place electronically and thus noiselessly and reliably, even during hail, snow and sub-zero temperatures. Even turbulent air and anabatic winds in the vicinity of the weather station are recorded
- **Temperature measurement**
- Heated **precipitation sensor** (1.2 watts): No false reports as a result of fog or dew. Dries quickly after precipitation has stopped
- Sending cycle for data 1 second

### Functions P03/3-RS485-GPS:

- Integrated **GPS receiver**. Output of UTC (Universal Time Coordinated), position (degree of longitude and latitude) and position of the sun (azimuth, elevation)

### Functions P03/3-RS485-CET:

- Integrated **GPS receiver**. Output of CET (Central European Time), automatic adjustment of daylight savings time.

### 1.0.1. Scope of delivery

- Weather station

## 1.1. Technical specifications

|                               |                                                                                 |
|-------------------------------|---------------------------------------------------------------------------------|
| Housing                       | Plastic material                                                                |
| Colour                        | White / translucent                                                             |
| Mounting                      | On-wall                                                                         |
| Degree of protection          | IP 44                                                                           |
| Dimensions                    | approx. 96 x 77 x 127 (W x H x D, mm)                                           |
| Weight                        | approx. 160 g                                                                   |
| Ambient temperature           | Operation -30...+50°C, Storage -30...+70°C                                      |
| Operating voltage             | 24 V DC ±10%.<br>A suitable power pack can be purchased from Elsner Elektronik. |
| Connection                    | Screw terminal                                                                  |
| Conductor cross-section       | Solid/fine-stranded conductors of 0.5...1.0 mm²                                 |
| Stripping length              | 6 mm                                                                            |
| Current                       | max. 130 mA, residual ripple 10%                                                |
| Data output                   | RS485                                                                           |
| Heating rain sensor           | approx. 1.2 W                                                                   |
| Measurement range temperature | -40...+80°C                                                                     |
| Measurement range wind        | 0...35 m/s                                                                      |
| Measurement range brightness  | 0 ... 99,000 lux                                                                |

The product conforms with the provisions of EU directives.

## 2. Installation and commissioning



Installation, testing, operational start-up and troubleshooting should only be performed by a qualified electrician.



**CAUTION!**  
**Live voltage!**

- Inspect the device for damage before installation. Only put undamaged devices into operation.
- Comply with the locally applicable directives, regulations and provisions for electrical installation.
- Immediately take the device or system out of service and secure it against unintentional switch-on if risk-free operation is no longer guaranteed.

Use the device exclusively for building automation and observe the operating instructions. Improper use, modifications to the device or failure to observe the operating instructions will invalidate any warranty or guarantee claims. Operate the device only as a fixed-site installation, i.e. only in assembled condition and after conclusion of all installation and operational start-up tasks, and only in the surroundings designated for it. Elsner Elektronik is not liable for any changes in norms and standards which may occur after publication of these operating instructions.

### 2.0.1. Installation position

Choose an installation position in the building where wind, rain and sun can be measured unhindered by the sensors. The weather station must not be installed underneath any structural parts from which water can still drip onto the rain sensor after it has stopped raining or snowing. The weather station must not be shaded by anything, such as building structures or trees.

At least 60 cm of clearance must be left all round the weather station. This facilitates correct wind speed measurement without eddies. The distance concurrently prevents spray (raindrops hitting the device) or snow (snow penetration) from impairing the measurement. It also does not allow birds to bite it. Please take note that an extended awning does not shade the device from sun and wind.

Temperature measurements can also be affected by external influences such as by warming or cooling of the building structure on which the sensor is mounted, (sunlight, heating or cold water pipes).

Magnetic fields, transmitters and interfering fields from electricity consumers (e.g. fluorescent lamps, neon signs, switched-mode power supplies etc.) can interfere with or even cut out reception of the GPS signal.

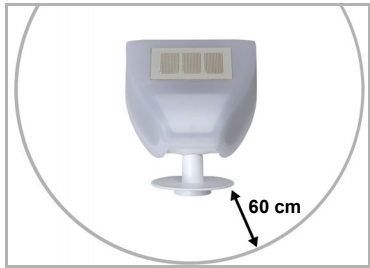


Fig. 1  
There must be at least 60 cm of space below, to the sides and in front of the weather station left from other elements (structures, construction parts, etc.).

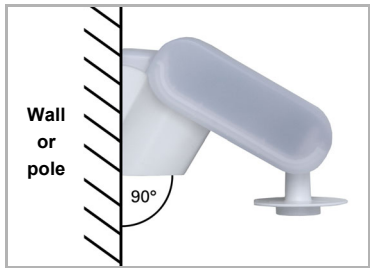


Fig. 2  
The weather station must be mounted on a vertical wall (or a pole).



Fig. 3  
The weather station must be mounted in the horizontal transverse direction (horizontally).

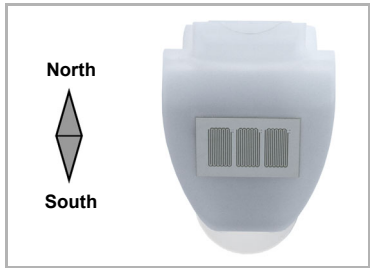


Fig. 4  
For installation in the northern hemisphere, the weather station must be aligned to face south.

For installation in the southern hemisphere, the weather station must be aligned to face north.

### 2.1. Mounting the sensor

#### 2.1.1. Attaching the mount

The sensor comes with a combination wall/pole mount. The mount comes adhered by adhesive strips to the rear side of the housing. Fasten the mount vertically onto the wall or pole.

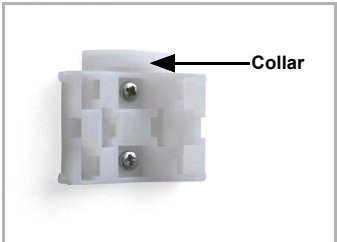


Fig. 5  
When wall mounting: flat side on wall, crescent-shaped collar upward.

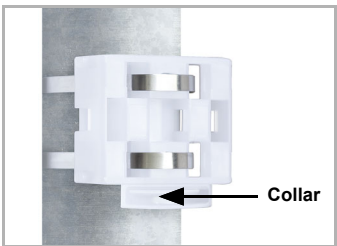


Fig. 6  
When pole mounting: curved side on pole, collar downward.

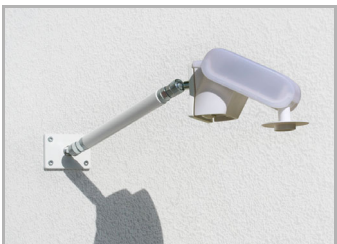


Fig. 7  
Different mounting arms are available from Elsner Elektronik as additional, optional accessories for flexible installation of the weather station on a wall, pole or beam (pictures of sensors exemplary). Example of the use of a mounting arm: Due to flexible ball joints, the sensor can be brought into ideal position.



Fig. 8  
Example use of the hinge arm mounting: With the hinge arm mounting, the weather station projects from beneath the roof overhang. Sun, wind and precipitation can act upon the sensors without hindrance.

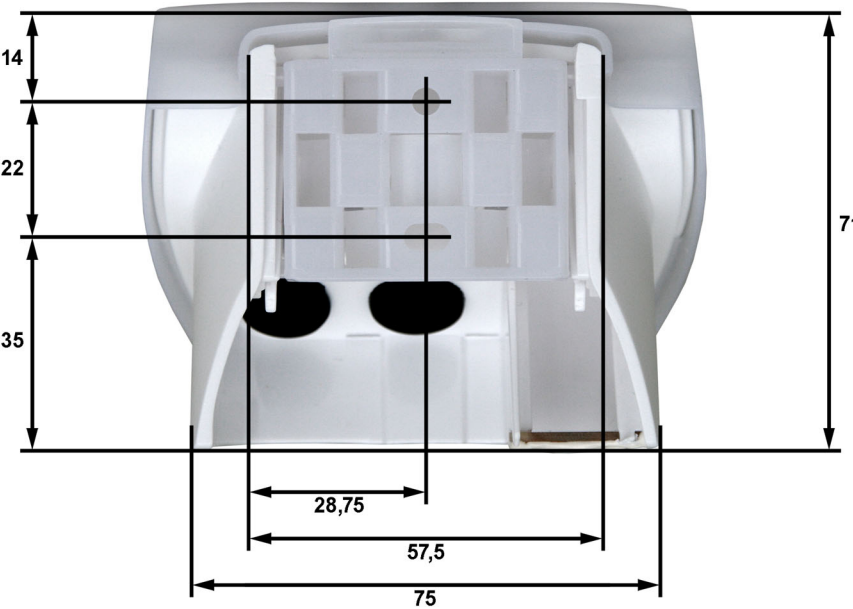
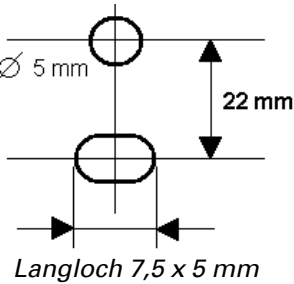


Fig. 9  
Example use of the hinge arm mounting: Fitting to a pole with worm drive hose clips

#### 2.1.2. View of rear side and drill hole plan

Fig. 10 a+b  
Drill hole plan

Dimensions of rear side of housing with bracket. Subject to change for technical enhancement.



2.1.3. Connection

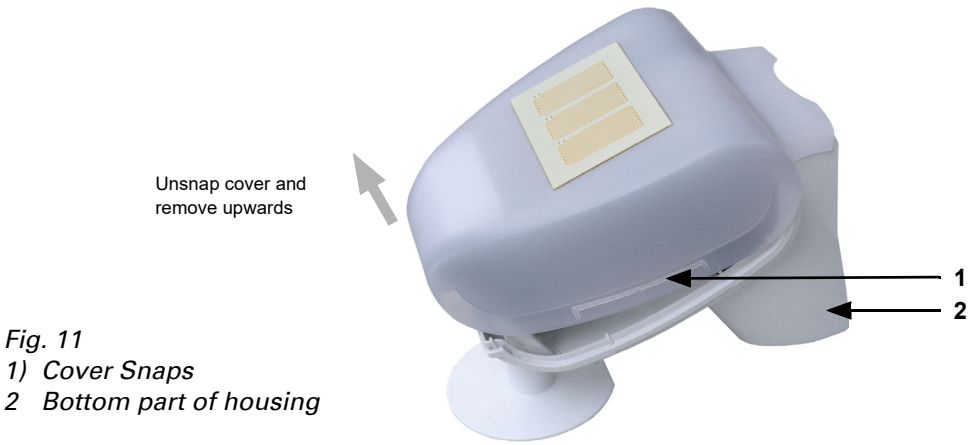


Fig. 11  
1) Cover Snaps  
2 Bottom part of housing

The weather station cover with the rain sensor snaps in on the left and right along the bottom edge (see figure). Remove the weather station cover. Proceed carefully, so as **not to pull off the wire** connecting the PCB in the bottom part with the rain sensor in the cover (wire with push-connector).

Connect the Modbus data line to terminals A and B. Connect the power supply (24 V DC) to terminals 1 and 2. Ensure correct connection!

Push the connecting cable through the rubber seal on the bottom of the weather station and connect the power and bus cables to the terminals provided for this purpose.

The connection is by typical telephone cable (J-Y(ST)Y 2 × 2 × 0.8).

The connection cable must be plugged in between the cover and circuit board.

2.1.4. PCB Layout

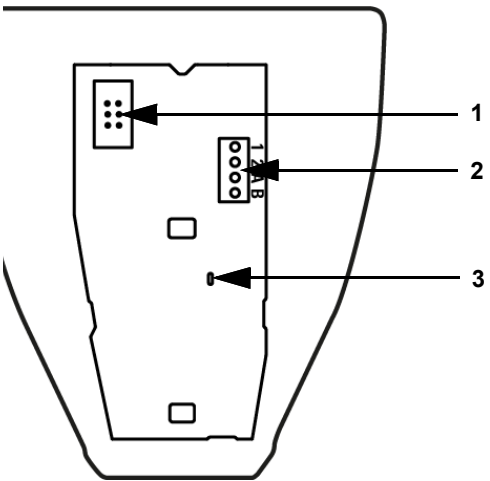


Fig. 12: Overview PCB  
1 Connecting cables to rain sensor in housing cover  
2 Terminal for connection  
1: +24 V DC | 2: „-“  
A: data | B: data  
3 GPS reception control LED

2.1.5. Mounting the weather station

Close the housing by putting the cover back over the bottom part. The cover must snap in on the left and right with a definite “click”.

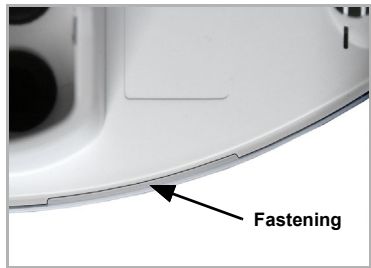


Fig. 13  
Make sure the cover and bottom part are properly snapped together! This picture is looking at the closed sensor from underneath.



Fig. 14  
Push the housing from above into the fastened mount. The bumps on the mount must snap into the rails in the housing.

To remove it, the weather station can be simply pulled upwards out of the mount, against the resistance of the fastening.



Fig. 15  
After installation, remove the protective sticker on the wind sensor and the “distance” sticker on the top of the cover.

2.2. Notes on mounting and commissioning

Do not open weather station if water (rain) might ingress: even some drops might damage the electronic system.

Observe the correct connections. Incorrect connections may destroy the weather station or connected electronic devices.

Please take care not to damage the temperature sensor (small blank at the bottom part of the housing.) when mounting the weather station. Please also take care not

to break away or bend the cable connection between the blank and the rain sensor when connecting the weather station.

The correct wind value may only be supplied about 10 seconds after the supply voltage has been connected.

2.3. Maintenance of the weather station

**WARNING!**  
**Risk of injury caused by components moved automatically!**  
The automatic control can start system components and place people in danger.

- Always isolate the system from the mains for servicing and cleaning.

The device must regularly be checked for dirt twice a year and cleaned if necessary. In case of severe dirt, the sensor may not work properly anymore.

**ATTENTION**  
The device can be damaged if water penetrates the housing.

- Do not clean with high pressure cleaners or steam jets.

3. Transfer protocol

All characters and/or digits are based on the ASCII standard, i.e. every reading processed internally as an integer or float value will always be broken down into and transferred in its individual ASCII format characters. They must then be reassembled in the reverse process by the receiver.

| Transfer rate | Data bits | Stop bit | Parity |
|---------------|-----------|----------|--------|
| 19200 Baud    | 8         | 1        | none   |

The checksum is calculated along by the receiver by adding all received bytes up until byte 35 and then compared with the checksum transferred from the P03.

Units:

| Temperatures       | Sun intensity | Daylight | Wind                 |
|--------------------|---------------|----------|----------------------|
| in degrees Celsius | in kilolux    | in Lux   | in metres per second |

P03/3-RS485-GPS:

| Byte No | Char        | Description                                                          |
|---------|-------------|----------------------------------------------------------------------|
| 1       | G           | Start of string                                                      |
| 2       | + / -       | Outdoor temperature in °C, sign                                      |
| 3       | 0 ... 9     | Outdoor temperature in °C, tens digit                                |
| 4       | 0 ... 9     | Outdoor temperature in °C, units digit                               |
| 5       | .           | Outdoor temperature in °C, decimal point                             |
| 6       | 0 ... 9     | Outdoor temperature in °C, tenths digit                              |
| 7       | 0 ... 9     | Sun south in kLux, tens digit                                        |
| 8       | 0 ... 9     | Sun south in kLux, units digit                                       |
| 9       | 0 ... 9     | Sun west in kLux, tens digit                                         |
| 10      | 0 ... 9     | Sun west in kLux, units digit                                        |
| 11      | 0 ... 9     | Sun east in kLux, tens digit                                         |
| 12      | 0 ... 9     | Sun east in kLux, units digit                                        |
| 13      | J / N       | Lux twilight indication                                              |
| 14      | 0 ... 9     | Daylight in Lux, hundreds digit                                      |
| 15      | 0 ... 9     | Daylight in Lux, tens digit                                          |
| 16      | 0 ... 9     | Daylight in Lux, units digit                                         |
| 17      | 0 ... 9     | Wind in m/s, tens digit                                              |
| 18      | 0 ... 9     | Wind in m/s, units digit                                             |
| 19      | .           | Wind in m/s, decimal point                                           |
| 20      | 0 ... 9     | Wind in m/s, tenths digit                                            |
| 21      | J / N       | Rain indication                                                      |
| 22      | ? / 1 ... 7 | UTC weekday (1 = mondy ... 7 = sunday   ? = UTC not OK)              |
| 23      | 0 ... 9     | UTC date day, tens digit                                             |
| 24      | 0 ... 9     | UTC date day, units digit                                            |
| 25      | 0 ... 9     | UTC date month, tens digit                                           |
| 26      | 0 ... 9     | UTC date month, units digit                                          |
| 27      | 0 ... 9     | UTC date year, tens digit                                            |
| 28      | 0 ... 9     | UTC date year, units digit                                           |
| 29      | 0 ... 9     | UTC time hour, tens digit                                            |
| 30      | 0 ... 9     | UTC time hour, units digit                                           |
| 31      | 0 ... 9     | UTC time minute, tens digit                                          |
| 32      | 0 ... 9     | UTC time minute, units digit                                         |
| 33      | 0 ... 9     | UTC time second, tens digit                                          |
| 34      | 0 ... 9     | UTC time second, units digit                                         |
| 35      | 0 / 1       | GPS report azimuth/elevation/longitude/latitude (1 = OK, 0 = not OK) |
| 36      | 0 ... 3     | Azimuth in °, hundreds digit                                         |
| 37      | 0 ... 9     | Azimuth in °, tens digit                                             |
| 38      | 0 ... 9     | Azimuth in °, units digit                                            |
| 39      | .           | Azimuth in °, decimal point                                          |
| 40      | 0 ... 9     | Azimuth in °, tenths digit                                           |
| 41      | + / -       | Elevation in °, sign                                                 |
| 42      | 0 ... 9     | Elevation in °, tens digit                                           |
| 43      | 0 ... 9     | Elevation in °, units digit                                          |
| 44      | .           | Elevation in °, decimal point                                        |
| 45      | 0 ... 9     | Elevation in °, tenths digit                                         |
| 46      | O / W       | Longitude in ° (O = east, W = west)                                  |

| Byte No | Char    | Description                          |
|---------|---------|--------------------------------------|
| 47      | 0 / 1   | Longitude in °, hundreds digit       |
| 48      | 0 ... 9 | Longitude in °, tens digit           |
| 49      | 0 ... 9 | Longitude in °, units digit          |
| 50      | .       | Longitude in °, decimal point        |
| 51      | 0 ... 9 | Longitude in °, tenths digit         |
| 52      | N / S   | Latitude in ° (N = north, S = south) |
| 53      | 0 ... 9 | Latitude in °, tens digit            |
| 54      | 0 ... 9 | Latitude in °, units digit           |
| 55      | .       | Latitude in °, decimal point         |
| 56      | 0 ... 9 | Latitude in °, tenths digit          |
| 57      | 0 ... 9 | Checksum thousands digit             |
| 58      | 0 ... 9 | Checksum hundreds digit              |
| 59      | 0 ... 9 | Checksum tens digit                  |
| 60      | 0 ... 9 | Checksum units digit                 |
| 61      | 0x03    | end identifier                       |

P03/3-RS485-CET:

| Byte No | char                  | Description                                        |          |
|---------|-----------------------|----------------------------------------------------|----------|
| 1       | W                     | Start of Weather Data                              |          |
| 2       | AT: sign              | Outdoor temperature sign - / +                     | in °C    |
| 3       | AT: 1st digit         | Outdoor temperature 1st digit (tens)               |          |
| 4       | AT: 2nd digit         | Outdoor temperature 2nd digit (units)              |          |
| 5       | AT: decimal point     | Outdoor temperature decimal point                  |          |
| 6       | AT: 3rd digit         | Outdoor temperature 3rd digit (tenths)             |          |
| 7       | SoS: 1st digit        | Sun south 1st digit (tens)                         | 1-99 klx |
| 8       | SoS: 2nd digit        | Sun south 2nd digit (units)                        |          |
| 9       | SoW: 1st digit        | Sun west 1st digit (tens)                          |          |
| 10      | SoW: 2nd digit        | Sun west 2nd digit (units)                         |          |
| 11      | SoO: 1st digit        | Sun east 1st digit (tens)                          |          |
| 12      | SoO: 2nd digit        | Sun east 2nd digit (units)                         |          |
| 13      | Twilight              | Twilight: J = Yes; N = No                          | < 10 lx  |
| 14      | Daylight 0-999Lx      | Daylight 1st digit (hundreds)                      | 0-999 lx |
| 15      | Daylight 0-999Lx      | Daylight 2nd digit (tens)                          |          |
| 16      | Daylight 0-999Lx      | Daylight 3rd digit (units)                         |          |
| 17      | Wind: 1st digit       | Wind 1st digit (tens)                              | in m/s   |
| 18      | Wind: 2nd digit       | Wind 2nd digit (units)                             |          |
| 19      | Wind: decimal point   | Wind decimal point                                 |          |
| 20      | Wind: 3rd digit       | Wind 3rd digit (tenths)                            |          |
| 21      | Rain                  | Rain: J = Yes; N = No                              |          |
| 22      | Week day: 1-7         | Week day:1 = Mon, 7 = Sun                          |          |
| 23      | Date: Day 1st digit   | Date: Day 1st digit (tens)                         |          |
| 24      | Date: Day 2nd digit   | Date: Day 2nd digit (units)                        |          |
| 25      | Date: Month 1st digit | Date: Month 1st digit (tens)                       |          |
| 26      | Date: Month 2nd digit | Date: Month 2nd digit (units)                      |          |
| 27      | Date: Year 1st digit  | Date: Year 1st digit (tens)                        |          |
| 28      | Date: Year 2nd digit  | Date: Year 2nd digit (units)                       |          |
| 29      | Time: Hour 1st digit  | Time: Hour 1st digit (tens)                        |          |
| 30      | Time: Hour 2nd digit  | Time: Hour 2nd digit (units)                       |          |
| 31      | Time: Min. 1st digit  | Time: Min. 1st digit (tens)                        |          |
| 32      | Time: Min. 2nd digit  | Time: Min. 2nd digit (units)                       |          |
| 33      | Time: Sec. 1st digit  | Time: Sec. 1st digit (tens)                        |          |
| 34      | Time: Sec. 2nd digit  | Time: Sec. 2nd digit (units)                       |          |
| 35      | Summer time           | J = Summer time N = Winter time<br>? = not defined |          |
| 36      | Checksum: 1st digit   | Checksum 1st digit (thousands)                     |          |
| 37      | Checksum: 2nd digit   | Checksum 2nd digit (hundreds)                      |          |
| 38      | Checksum: 3rd digit   | Checksum 3rd digit (tens)                          |          |
| 39      | Checksum: 4th digit   | Checksum 4th digit (units)                         |          |
| 40      | End identifier        | end identifier 0x03                                |          |

4. Disposal

Observe legal regulations and do not dispose of with household waste!